Aggregate biocapacity does not identify its composition: a two-land model of ecological-capital drawdown

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Abstract

Aggregate biocapacity is a value-weighted total over heterogeneous accounts; the total alone does not identify their composition. We formulate a two-land delay-differential model in which account shortfall triggers one-way conversion from ecological-capital to fast-provisioning area, while the capital-quality state q evolves separately. The conversion contribution to aggregate biocapacity is positive when fast-land yield exceeds the effective per-hectare value of capital land. In a reduced frozen-demand equilibrium relation with $q = q_{\max}$, debt and technology wave off, and restoration disabled, the balance point is unique and the floor collision has no fold or critical slowing down. This branch result does not eliminate trajectory dependence: conversion without restoration leaves realized A_c below its pre-overshoot level after conversion stops, but does not imply perpetual decline at equilibrium. In a representative conditional run, A_c falls 46.2%, $\Upsilon(q)$ falls 10.3%, and the modeled capital contribution $\Upsilon(q)A_c$ falls 51.7% while aggregate B rises 39%; these are model outputs, not empirical stock estimates. In Edition 2026 of the National Footprint and Biocapacity Accounts, world biocapacity rises 24.9% from 1961 to 2023 while B/P falls 52.7% and the global production footprint per person rises 16.7%. Cropland contributes +24.06 percentage points to the total change, Forest biocapacity −5.95, and built-up land +7.44 as an accounting class, not ecological capital. NFA component values are accounting signals, not measurements of biomass, soil condition, wetland extent, fish biomass, or A_c. No common-boundary global physical-stock series for the relevant components was assembled; accordingly, no physical-stock composite or decline finding is reported. The physical-stock scoreboard is an unpopulated specification; heterogeneous measures require declared boundaries, valuation or normalization, and substitution rules.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation, identifiability, critical slowing down, delay differential equations, carrying capacity

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1. Introduction

Carrying capacity is central to ecology but difficult to estimate. Classical population models treat it as a fixed ceiling, and global human-ecology models often represent it as a static upper bound on population (Cohen, 1995). Such formulations can omit feedbacks through which overshoot degrades a resource base and changes future capacity. A related problem concerns the information contained in aggregate measures of regenerative supply.

The National Footprint and Biocapacity Accounts (NFA) report annual biocapacity and Ecological Footprint values in global hectares (gha). A global hectare is a hectare normalized to world-average biological productivity for a specified year; the annual basis of the account is implicit in the unit (Global Footprint Network, 2009, 2019). Biocapacity represents annual regenerative supply, and the Ecological Footprint represents annual demand on biologically productive area. Because yield and equivalence factors are embedded in this productivity-normalized area, the aggregate does not distinguish changes in productivity, total area, and land-class composition. Composite-indicator analyses show that additive aggregation embeds trade-offs, allowing a deficit in one component to be offset by a surplus in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022); related critiques address the interpretation of the Ecological Footprint (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016). Aggregate biocapacity remains informative as a summary measure, but its interpretation depends on the component values and the processes that generate them.

Coupled human–environment models have represented environmental change through single stocks or aggregate balances, including the World3 system (Meadows et al., 1972, 1974, 2004), Ricardo–Malthus models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and bidirectionally coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Estimates of Earth's carrying capacity span several orders of magnitude (Cohen, 1995; van den Bergh and Rietveld, 2004); other work questions the concept for human populations (Simon, 1996), models a variable ceiling (Meyer, 1994; Meyer and Ausubel, 1999), or treats natural capital as a depreciating asset (Dasgupta, 2021). Demographic projections used in global assessments commonly specify population independently of environmental constraints (United Nations, 2022). The land-use literature documents conversion of forest, grassland, and wetland to cropland (Haberl et al., 2007; Krausmann et al., 2013;

Erb et al., 2017). These transitions motivate analysis of how aggregate biocapacity responds when land classes differ in yield and regeneration dynamics.

The model distinguishes fast-provisioning land from ecological-capital land and reserve area. Capital-land quality is represented by a separate state, while demand is compared with the human-available annual account total. A shortfall may induce conversion from capital land to fast-provisioning land; when the fast-land yield exceeds the effective per-hectare capital value, conversion contributes positively to aggregate biocapacity even as capital-land area A_c declines. This is a conditional model mechanism, not an empirical interpretation of NFA component movements. Related material-ledger analysis discusses productivity and aggregation effects in a broader sustainability framework (Abaee, 2026c).

The analysis addresses three questions. (1) Under what conditions can conversion raise aggregate biocapacity while capital-land area A_c falls, how long can this signature persist, and can the composition be identified from the aggregate? (2) Does the resulting drawdown involve a catastrophic shift, fold, hysteresis, or early-warning signal, or does it arise through a different structure? (3) Under what conditions is the drawdown self-limiting, and what mechanisms permit recovery?

The model analysis derives the sign condition for the conversion contribution, an identifiability result for aggregate-only observations, and the frozen-demand balance relation and its stability properties. Numerical simulations examine transient drawdown, debt accumulation, and restoration under representative parameter values. The empirical analysis uses NFA Edition 2026 world accounts for 1961–2023 and a same-edition 1961–2022 sensitivity; these account results are kept separate from physical-stock evidence. Model parameters are representative rather than fitted, so the trajectories characterize conditional mechanisms rather than forecasts.

2. Methods

2.1. A two-land model with conserved area and a separate quality state

The bioproductive area A_{tot} is divided among three land classes: fast-provisioning land A_f (cropland and intensively managed systems), ecological-capital land A_c (forest, soil, wetland, and long-rotation systems, represented as one effective class), and reserve land A_r (fallow, retired, or not yet allocated land), with $A_f + A_c + A_r = A_{tot}$ at all times. Capital land carries a per-hectare

quality index $q \in [0, q_{\max}]$ (a capacity state, not an area), which regenerates logistically,

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

so that the per-hectare value of capital land, in annual NFA-account-yield units, is

$$\Upsilon_c(q) = b_c + b_{G,c} \, G_c(q), \tag{2}$$

the sum of a direct per-hectare yield b_c and the regeneration-value coefficient $b_{G,c}$ times the quality growth $G_c(q)$. Biocapacity is the two-class sum

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q) \, A_c, \tag{3}$$

in gha, with the annual accounting basis implicit in the unit (Global Footprint Network, 2009, 2019). Demand is the footprint $E = eP$ of a population P with constant per-capita footprint e (gha cap^{-1}), and the shortfall above the human-available amount is

$$S = [E - \sigma_f Y_f - \sigma_c Y_c]_+ , \tag{4}$$

with $[x]_+ = \max(x, 0)$ exact (never smoothed) and $\sigma_f = \sigma_c = 1$ in the baseline. The shortfall is closed by converting capital land into fast land at a finite rate,

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\left(\frac{S}{b_f}, \; A_c - A_c^{\min}\right) \qquad [\text{ha yr}^{-1}], \tag{5}$$

where S/b_f is the area that would close the shortfall at the fast-land yield, $A_c^{\min}$ is a typed viability floor, and τ_{conv} is the conversion time constant. The land areas evolve through one-to-one transfers,

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

where μA_r is cropland maintenance (replanting from the reserve), $\eta_f A_f$ is fast-land retirement to the reserve, and R_{rc} is an explicit, surplus-gated restoration flow from reserve to capital land,

$$R_{rc} = \chi_r \, A_r \, [B - E]_+ , \tag{9}$$

which is the only route by which area returns to the capital-land class. Equations (6)–(8) sum to zero, so the reserve is a constrained balance rather than an independent state. Quality regenerates with a recruitment delay; population follows a delayed logistic law (Pearl, 1925; Hutchinson, 1948) with carrying capacity $K = \max(B/e, K_{\min})$; degradation acts on the surviving fast-land yield through the leaky debt state D:

$$\dot{q} = G_c\big(q(t-\tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t-\tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E-B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\,\frac{\Delta b}{1+e^{-\kappa_w(t-t_{wave})}}, \qquad \omega_{Tb} \in \{0,1\}. \tag{14}$$

Because model time t is measured in years and $E-B$ is an annual account amount in gha, the leaky shortfall accumulator D has units gha yr: both $\dot{D}$ and ηD have units gha. This is a model-internal time-integrated liability, not an NFA-reported footprint or biocapacity item.

The technology wave T_b is a bounded logistic yield step controlled by ω_{Tb}: $\omega_{Tb} = 0$ in the baseline, main figures, and all tables except the matched experiment in SI §S7, where $\omega_{Tb} = 1$. The state is (A_f, A_c, q, P, D) with one delay each in regeneration and demography; a timber-liquidation channel and a direct fast-to-capital restoration flow are stated extensions that remain off in every reported run (Supplementary Information, SI §S1).

The positive-part operator is exact, so $u_c = 0$ whenever $S = 0$. Conversion is finite-rate: S/b_f is an area and enters u_c only after division by τ_{conv}. Equations (6)–(8) conserve area through one-to-one transfers; q changes the effective per-hectare value without transferring area. The maintenance term μA_r can sustain a positive equilibrium value of A_f without conversion.

The model contains two positive feedbacks. In the conversion feedback, when $b_f > \Upsilon_c(q)$, conversion from A_c to A_f increases B at the conversion margin. The resulting increase in carrying capacity $K = B/e$ can support population growth, raise subsequent demand, and increase the shortfall that drives conversion. Other contributions to $\dot{B}$ determine its net sign. In the debt feedback, a shortfall increases D while debt repayment reduces it; rising D lowers b_f, reduces B, and can thereby increase the shortfall. Under

fixed demand with a sustained shortfall, this leaky state approaches a finite equilibrium.

2.2. Parameters and their status

Table 1 lists the baseline parameter values and their evidential status. Ecological-capital land aggregates heterogeneous components that differ in extent, condition, stock, yield, and regeneration time; the model represents them as one effective class. Most parameter values are therefore representative rather than fitted. Independent recovery observations are system-specific: forest biomass has median $t_{50} \approx 30\,\mathrm{yr}$, collapsed fish stocks $t_{50} \approx 5.6\,\mathrm{yr}$, and soil organic carbon re-equilibrates over approximately 17–23 yr (SI §S5.5) (Poorter et al., 2016; Hutchings and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011). These observations provide timescale context for the effective capital-land representation; they do not identify a pooled global rate, calibrate ρ_c, or determine τ_g. The NFA cropland biocapacity-per-area quotient and the supplementary FAOSTAT cereal-yield/cropland-area proxy are descriptive account diagnostics, not fitted values of model b_f or physical observations of ecological capital (SI §S5). The proxy uses an explicit 1961 NFA account anchor, but its area and yield boundaries do not establish a mapping to the model's fast-provisioning land. The baseline lies in the composition-premium regime, $\Delta b_{\mathrm{conv}} = b_f - \Upsilon_c(q) > 0$, which permits a positive conversion contribution; the baseline parameter set defines a conditional model regime rather than an empirical prediction.

2.3. Data and the observation operator

The empirical analysis uses the official National Ecological Footprint and Biocapacity Accounts, Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The primary window is the complete world series 1961–2023; rows for 2024–2025 are estimated and excluded. The 1961–2022 sensitivity uses the same 2026 edition and population series; both analysis windows therefore use a single data vintage. From the workbook's `world_data` sheet, B is `biocap_total_gha`; the empirical footprint series is `efp_total_gha`, the Global Ecological Footprint of Production, denoted E_{prod}; and population is `pop_1000_persons`, defined in the workbook as UN PopStat population. Per-capita quantities are recomputed by dividing each total account value by that population in persons, rather than using the rounded workbook per-capita columns. Ratios, indices, and logarithmic changes are dimensionless.

Table 1: Baseline parameter values and evidence status. Representative values are effective, lumped quantities; the NFA account/area quotient is descriptive, not a physical-yield measurement. Extension parameters are active only in scenarios specified in this table or the SI. Symbols and units are listed in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality regeneration rate of capital land	representative; not fitted
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (2026 NFA account value/area quotient ≈ 1.045 in 1961; not a physical-yield measure)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	normalized quality ceiling	normalization
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delays	representative; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under frozen demand
$A_c^{\min}$, $A_r^{\min}$, $K_{\min}$	0.05 ha, 0.10 ha, 0.02 cap	typed floors	domain constraints
Δb, κ_w, t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in the matched SI S7 experiment

The component series retain the workbook labels Cropland, Grazing land, Forest biocapacity, Fishing grounds, and Built-up land. These are NFA account values, not measurements of biomass, soil condition, wetland extent, fish biomass, or the model's effective area A_c. Built-up land is an NFA accounting class and is not ecological capital. Accordingly, the land-type movements are used as accounting-composition signals only. Variable definitions and endpoint calculations are summarized in SI §S5.

A supplementary proxy sensitivity uses OWID's FAOSTAT-derived series for world cropland area (arable land plus permanent crops) and cereal yield as separate physical inputs (Our World in Data, 2026a,b). The cereal series is a sentinel, not an all-crop yield measure or an NFA yield factor. Its base-year account level is anchored explicitly to 1961 NFA cropland biocapacity; the proxy $X_t = \hat{b}_{f,t} A_{f,t}$ and residual $C_t = B_t - X_t$ are therefore an assumed observation operator, not a measured partition of physical capital. C is an accounting residual only and cannot be identified with $\Upsilon(q)A_c$ without independent, boundary-consistent inputs and a declared mapping and valuation. The proxy and its calibration sensitivity are reported in SI §S5; no physical-stock trend is inferred from it.

The aggregate-versus-per-capita comparison uses the exact identity $\Delta \ln B = \Delta \ln(B/P) + \Delta \ln P$. This extensive/intensive split is an accounting identity, not a causal decomposition of biocapacity production.

2.4. Numerical solution

The delay system is integrated by the method of steps with an explicit first-order Euler scheme. The demonstration run and threshold bisection use $\Delta t = 0.05\,\mathrm{yr}$; the drawdown-depth sweeps use $\Delta t = 0.1\,\mathrm{yr}$, with selected thresholds checked at $0.05\,\mathrm{yr}$. The positive-part operator is exact, and state bounds enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, and $K \geq K_{\min}$. Pre-run histories of B and q are constant at the initial state over the maximum active delay. In the controlled wave experiment, both runs use the common no-wave prehistory; the active $T_b(0)$ changes current $B(0)$, and subsequent wave-adjusted B values enter the delayed history used in $K(t-\tau_p)$. The operational floor-proximity thresholds depend on tolerance and step size and are therefore reported as bands. The delayed-quality specification $G_c(q(t-\tau_g))$ is used for the reported trajectories; results under the current-quality specification $G_c(q(t))$ are compared in SI §S3. The matched technology-wave experiment and step-size sensitivity are reported in SI §S7.

3. Results

3.1. Composition masking and aggregate non-identifiability

The per-hectare value of capital land $\Upsilon_c(q)$ is a concave quadratic in q with an interior maximum at $q^* = q_{\max}/2$, where $\Upsilon_c(q^*) = b_c + b_{G,c}\rho_c q_{\max}/4 = 0.066\,\mathrm{gha\,ha^{-1}}$ at the baseline; the regenerative term there ($0.016\,\mathrm{gha\,ha^{-1}}$) is about one third of the direct yield ($0.05\,\mathrm{gha\,ha^{-1}}$; ratio 0.32), so capital value is only modestly enhanced by regeneration. The quantity that governs what conversion does to the aggregate is the *composition diagnostic*

$$\Delta b_{\mathrm{conv}}(q, D) = b_f(D) - \Upsilon_c(q), \tag{15}$$

the difference between the per-hectare account yield of fast land and the effective per-hectare value of capital land. At the baseline $\Delta b_{\mathrm{conv}} = +0.787,\ +0.784,\ +0.794\,\mathrm{gha\,ha^{-1}}$ at $q = 0.3, 0.6, 0.9$: the conversion contribution is positive at each of these baseline quality values, although other terms still determine the net account change. These sign cases exhaust the possibilities under the quality-based formulation: when $\Delta b_{\mathrm{conv}} > 0$, conversion contributes positively to B; when $\Delta b_{\mathrm{conv}} = 0$, its first-order contribution is zero; and when $\Delta b_{\mathrm{conv}} < 0$, conversion contributes negatively. At fixed q, Y_c is linear in A_c and has no interior maximum as a function of A_c.

Proposition 1 (Composition identity, mask and non-identifiability)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\mathrm{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{16}$$

(i) Whenever $u_c > 0$, the conversion contribution to $\dot{B}$ is $\Delta b_{\mathrm{conv}}(q)u_c$ and is positive if and only if $\Delta b_{\mathrm{conv}}(q) > 0$; with $R_{rc} = 0$, the same transfer gives $\dot{A}_c = -u_c < 0$. At any state where the maintenance–retirement imbalance and quality-growth terms vanish ($\mu A_r = \eta_f A_f$ and $\dot{q} = 0$), the net derivative is $\dot{B} = \Delta b_{\mathrm{conv}}(q)u_c$, so its sign has this same condition.

(ii) In the reduced frozen-demand, fixed-D case in which $\mu A_r = \eta_f A_f$ and $\dot{q} = 0$ throughout, and with $\sigma_f = \sigma_c = 1$, $\Delta b_{\mathrm{conv}} > 0$ implies $\dot{S} = -\dot{B} = -\Delta b_{\mathrm{conv}} u_c < 0$ whenever $S > 0$. Thus the shortfall decreases to zero if the balance is reached first; if the floor constraint becomes active first,

A_c approaches $A_c^{\min}$ asymptotically and the shortfall can persist. This monotonicity is not asserted when the other terms in the full model are nonzero.

(iii) At fixed $b_f > 0$ and fixed q (so that $\Upsilon_c(q) > 0$), the observation $B = b_f A_f + \Upsilon_c(q) A_c$ is one scalar constraint on two land areas. At an interior feasible state it leaves a local continuum of decompositions: a sufficiently small δA_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$, with $\delta A_r = -\delta A_f - \delta A_c$, while preserving B and the area total, provided the area bounds remain satisfied. Thus an aggregate-only observation does not locally identify A_c. For a finite observed change ΔB, the algebraic offset $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$ is feasible only when the resulting land areas respect their bounds; the admissible range depends on the state and need not be the whole interval from $A_c^{\min} - A_c(0)$ to zero. If b_f and A_f are independently observed, the residual $C = B - b_f A_f = \Upsilon_c(q) A_c$ identifies that product, not Υ_c and A_c separately. An independent observation of A_c would identify the effective value $\Upsilon_c(q) = C/A_c$; even then, b_c, $b_{G,c}$, ρ_c, and q are not separately identified from a single value. With q independently observed over a sufficiently informative range, $\Upsilon_c(q)$ can at most identify b_c and the product $b_{G,c}\rho_c$, not $b_{G,c}$ and ρ_c separately.

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c + \mu A_r - \eta_f A_f) - \Upsilon_c u_c + b_{G,c} G_c'(q) A_c \dot{q}$, which is (16). (i) Follows from (16) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) Under the reduced-case conditions in the statement, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ on $S > 0$. Since S is bounded below by zero and A_c is bounded below by $A_c^{\min}$, conversion either closes the shortfall or is limited by the floor. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C identifies only $\Upsilon_c(q) A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a shows the mask at the baseline under frozen demand $E = 0.825\,\text{gha}$ (restoration off, $A_{f0} = 0.6\,\text{ha}$, $A_{c0} = 1.5\,\text{ha}$, dimensionless $q_0 = 0.9$, $A_r(0) = 0.4\,\text{ha}$). Biocapacity rises from 0.594 to 0.825 gha (+39%) while capital land area A_c falls from 1.500 to 0.808 ha (−46%) and fast land A_f rises from 0.600 to 0.923 ha. The dimensionless quality state q rises from 0.9 to

1.0, but because $q_0 = 0.9 > q^* = q_{\max}/2 = 0.5$, this rise lies on the descending side of the concave per-hectare function $\Upsilon_c(q)$: its value falls from 0.05576 to 0.05000 gha ha^{-1}. Accordingly, the modeled capital contribution $\Upsilon_c(q)A_c$ to B falls from 0.08364 to 0.04038 gha (by 51.7%, about 52%), distinct from the 46% decline in capital-land area. The area changes satisfy $\Delta A_f + \Delta A_c + \Delta A_r = 0$, as required by land conservation. The trajectory has a rapid initial phase followed by a long asymptotic tail: because the deficit-side land subsystem is affine (Proposition 2(ii)), the approach to the balance point is exponential at the slow rate $|\lambda_1| = 0.054\,\mathrm{yr}^{-1}$ (e-folding ≈ 19 yr), so within 25 yr 98% of the rise in B and 85% of the fall in A_c have occurred ($B = 0.820$ gha, $A_c = 0.913$ ha; 97% of the fall by 50 yr), while the strict inequalities $\dot{B} > 0$ and $\dot{A}_c < 0$ persist for ≈ 149 yr at numerical resolution. The slow tail reflects reserve equilibration: retirement of fast land to the reserve lowers the fast-land contribution, reopens a small shortfall, and sustains further conversion until the reserve reaches its equilibrium share $A_r/A_f = \eta_f/\mu$. The window is therefore not a fixed horizon: its scale is set by λ_1, which depends on τ_{conv}, μ, η_f and the yield ratio b_c/b_f (SI §S2.2), and its amplitude by Δb_{conv} and the initial deficit.

Proposition 1(iii) constrains the empirical interpretation. An aggregate observation B alone cannot distinguish a composition mask from aggregate expansion. Independent proxies for fast-land area A_f and its per-hectare yield b_f allow the residual $C = B - b_f A_f$ to be calculated, but C identifies only the capital-land contribution unless capital-land area and its effective per-hectare value are independently observed. The term *composition mask* denotes a trajectory with B rising while A_c falls. Its conversion contribution is positive when $\Delta b_{\mathrm{conv}} > 0$, although the net sign of $\dot{B}$ also depends on the other terms in (16). A trajectory with stable E and declining A_c does not require a positive conversion contribution; the two patterns are distinguishable when account demand and capital-land area are both observed.

3.2. The frozen-demand balance branch ends at a floor collision, not a fold

With endogenous population there is no interior steady state with active conversion: at a demographic equilibrium $P = K = B/e$, so $E = B$ and $S = 0$. Conversion is therefore a transient regime, and its steady states exist only under a frozen exogenous demand pinned above the flow. The frozen-demand system provides a reduced setting in which the equilibrium response can be characterized.

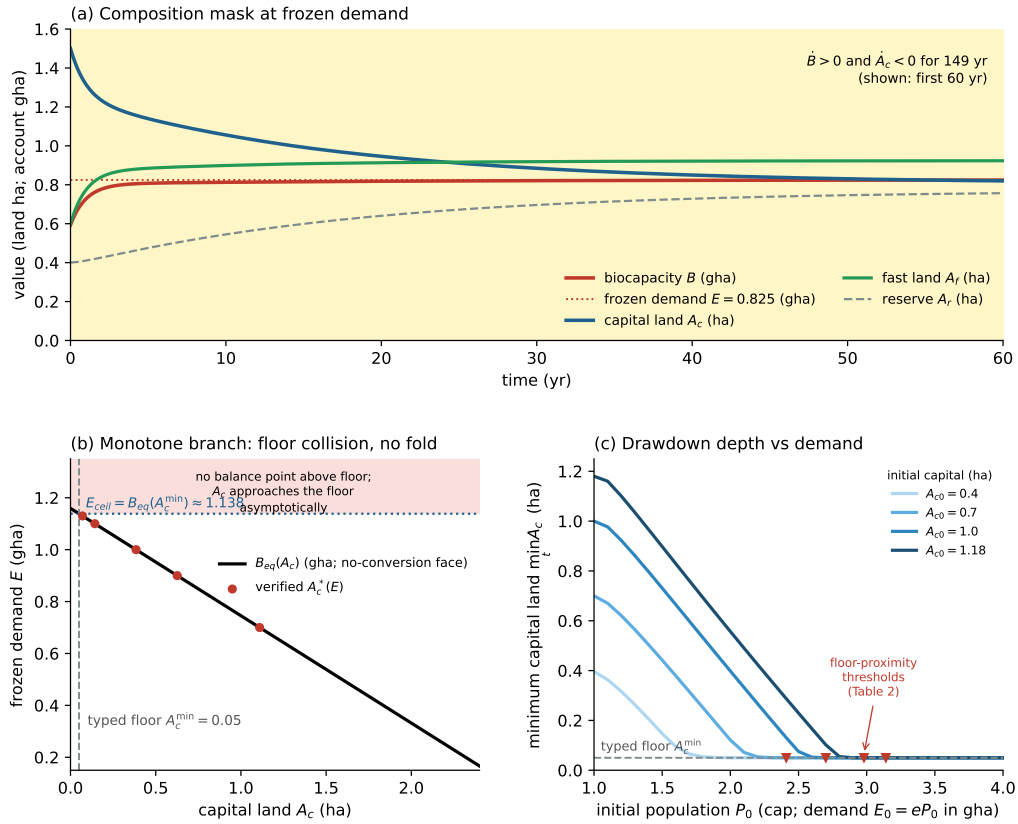


Figure 1: Model dynamics at the baseline parameter set (Table 1). Account amounts B and E are in gha; land areas A_f, A_c, A_r are in ha; population P_0 is in cap. (a) Composition mask under frozen demand $E = 0.825$ (restoration off, $A_{f0} = 0.6$, $A_{c0} = 1.5$, dimensionless $q_0 = 0.9$): biocapacity B rises from 0.594 to 0.825 while capital land A_c falls from 1.500 to 0.808 and fast land A_f rises from 0.600 to 0.923; the shaded region marks the interval on which $\dot{B} > 0$ and $\dot{A}_c < 0$ hold simultaneously (≈ 149 yr at numerical resolution; only the first 60 yr are shown; 98% of the rise in B and 85% of the fall in A_c occur within the first 25 yr). (b) The frozen-demand balance branch $B_{eq}(A_c)$ of Proposition 2, strictly decreasing in A_c, with the balance points calculated from Eq. (18) $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13$; the branch meets the typed floor $A_c^{\min} = 0.05$ at the demand ceiling $E_{\text{ceil}} \approx 1.138$, above which no admissible balance point exists and A_c approaches the floor (shaded). (c) Endogenous-demand runs ($\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$): the minimum capital land reached during a run is a continuous, monotone function of initial population P_0 (with initial demand $E_0 = eP_0$) for each initial capital A_{c0}; the floor is approached asymptotically, and the floor-proximity thresholds of Table 2 (triangles) are the operational points at which the floor is approached to within 10^{-6} ha. There is no discontinuity in the outcome as a function of demand.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *Assume frozen demand E, $T_b \equiv 0$, $R_{rc} \equiv 0$, $q = q_{\max}$ (the attractor of (10) for $q(0) > 0$) and $D = 0$. Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\,k\,(A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0}k - b_c)A_c. \tag{17}$$

(i) Equilibria of the land subsystem on the no-conversion face $S = 0$ satisfy $A_f = k(A_{tot} - A_c)$, $A_r = (1-k)(A_{tot} - A_c)$ and $B_{eq}(A_c) \geq E$. If $b_{f0}k > b_c$, B_{eq} is strictly decreasing, so the balance point $B_{eq}(A_c^*) = E$ is unique,

$$A_c^*(E) = \frac{b_{f0}kA_{tot} - E}{b_{f0}k - b_c}, \tag{18}$$

The algebraic root lies in the feasible land interval $A_c \in [A_c^{\min}, A_{tot}]$ exactly when $B_{eq}(A_{tot}) \leq E \leq E_{\text{ceil}} := B_{eq}(A_c^{\min})$; it lies strictly above $A_c^{\min}$ iff $E < E_{\text{ceil}}$. For $E < B_{eq}(A_{tot})$, the root lies above A_{tot} and feasible no-conversion equilibria already satisfy $B_{eq} \geq E$.

(ii) On the deficit side ($S > 0$, $S/b_f < A_c - A_c^{\min}$) the land subsystem is affine in $z = (A_f, A_c)$, $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}}\Big(k - \frac{b_c}{b_f}\Big), \tag{19}$$

and $\operatorname{tr} J < 0$. J does not depend on E or on the state, $\det J > 0$ exactly when B_{eq} is decreasing, and the fixed point of the affine system is the balance point of (i). For $B_{eq}(A_{tot}) \leq E < E_{\text{ceil}}$ the balance point lies above the typed floor and every trajectory that remains in this regime converges to it at the constant rates given by the eigenvalues of J; for $E > E_{\text{ceil}}$ the affine fixed point lies below the floor, so the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with a limiting persistent shortfall $S^* = E - E_{\text{ceil}} > 0$. The ceiling is a collision of this reduced equilibrium relation with a face of the admissible set, not a bifurcation of its vector field: there is no second branch or fold, no hysteresis of this frozen-demand equilibrium relation, and the leading eigenvalue does not tend to zero as $E \to E_{\text{ceil}}$. These statements apply under the assumptions above; they are not claims that the realized land allocation is independent of its history.

(iii) Beyond the ceiling, with degradation active, the debt converges to the unique root $D^*(E)$ of $\eta D = E - B_{eq}(A_c^{\min}; D)$, where $B_{eq}(\cdot\,; D)$ uses $b_{f0}e^{-\alpha D}$, provided $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; D^* is increasing in E.

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r/A_f = \eta_f/\mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3) with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \geq E$. Strict monotonicity and (18) follow from $b_{f0}k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\text{ceil}}$, tends to $+\infty$, and is strictly increasing when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \geq -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

The uniqueness result is about the reduced equilibrium relation, not path independence of realized states. In the baseline trajectory, $R_{rc} = 0$ and the conversion flow u_c has no reverse counterpart. When the shortfall closes, A_c stops declining and approaches its lower balance value; it remains below its pre-overshoot level because this run contains no restoration flow. The quality state q evolves separately and can continue changing $\Upsilon_c(q)$ even as A_c settles. Thus the model has a unique frozen-demand balance point under Proposition 2's restrictions, but a history-dependent post-overshoot land allocation; this is not perpetual decline at equilibrium and is not a general claim about real ecosystems.

At the baseline $k = 0.5455$, the slope $b_{f0}k - b_c = 0.414\,\text{gha}\,\text{ha}^{-1}$, and $E_{\text{ceil}} = 1.138\,\text{gha}$; the balance points $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070\,\text{ha}$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13\,\text{gha}$ agree with the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative, $\lambda_1 = -0.0537\,\text{yr}^{-1}$ and $\lambda_2 = -0.998\,\text{yr}^{-1}$; λ_1 varies only weakly with the fast-land yield, from $-0.0522\,\text{yr}^{-1}$ at $b_f = 0.60$ to $-0.0556\,\text{yr}^{-1}$ at $b_f = 2.00$ (Table 2a), and never changes sign. Beyond the ceiling the debt fixed points are $D^* = 1.19$ and $5.38\,\text{gha}\,\text{yr}$ at $E = 1.158$ and $1.238\,\text{gha}$ (integrated values 1.21 and $5.39\,\text{gha}\,\text{yr}$); the monotonicity condition of (iii) holds with margin ($\alpha b_{f0} k (A_{tot} - A_c^{\min}) = 0.034\,\text{yr}^{-1} < \eta = 0.05\,\text{yr}^{-1}$), so under frozen demand the debt saturates rather than running away.

Here, critical slowing down denotes the case in which the least-negative eigenvalue of the relevant linearization tends to 0^- as a parameter approaches a fold, so that the return time $T_{ret} \approx -1/\lambda_1$ diverges. In Proposition 2 the linearization is constant along the entire branch, so neither λ_1 nor $T_{ret} \approx 19\,\mathrm{yr}$ changes as $E \to E_{\mathrm{ceil}}$. The corresponding model signal is the approach of A_c to its typed floor, which requires an observation of land-class composition rather than the aggregate alone.

Under endogenous demand, the minimum capital-land area is a continuous, monotone function of initial population P_0 for each initial value A_{c0} (Fig. 1c). The floor is approached asymptotically because $u_c \leq (A_c - A_c^{\min})/\tau_{\mathrm{conv}}$, and higher initial capital-land area supports higher demand before floor proximity. Table 2b reports operational proximity thresholds for distances of 10^{-2}, 10^{-4}, and $10^{-6}\,\mathrm{ha}$; these tolerance-based values do not mark discontinuities. At the $10^{-6}\,\mathrm{ha}$ tolerance, $P_0 \approx 2.41, 2.70, 2.98, 3.14\,\mathrm{cap}$ for $A_{c0} = 0.40, 0.70, 1.00, 1.18\,\mathrm{ha}$, respectively. The thresholds increase with initial capital-land area, and none of the tested tolerances produces a jump. At $A_{c0} = 1.0\,\mathrm{ha}$, the threshold is identical to three decimal places for $\tau_g = 5$, 20, and $40\,\mathrm{yr}$. It is also identical to three decimal places with $\chi_r = 0.10$ and $\chi_r = 0\,\mathrm{gha^{-1}\,yr^{-1}}$ (SI §S4). The delay sensitivity is small because regeneration contributes a modest share of capital-land value in the baseline; surplus-gated restoration does not affect the shortfall-driven drawdown because it is inactive while $B < E$.

3.3. The mask is bounded: technology gain is parameter-bounded, debt erosion is shortfall-bounded

Both the technology wave and the degradation act on the same coefficient, the fast-land yield b_f of (12), and they do so multiplicatively. This yields a structural asymmetry.

Proposition 3 (Boundedness asymmetry)**.** *Under* (12)–(14)*:*

(i) the technology factor satisfies $1 \leq (b_{f0}+T_b(t))/b_{f0} \leq 1+\Delta b/b_{f0}$ *for all* t*, so the logarithmic gain of the yield channel is bounded by* $\ln(1+\Delta b/b_{f0})$ *uniformly in time (*0.78 *at the baseline);*

(ii) the debt satisfies $D(t) \leq \max\{D(0),\ \sup_{s\leq t}[E-B]_+(s)/\eta\}$*, and* $D \to S^*/\eta$ *whenever the shortfall settles at* S^**; the degradation factor* $e^{-\alpha D}$ *is therefore bounded below only through the sustained shortfall;*

Table 2: Model results at the baseline parameter set. (a) Composition diagnostic at $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066\,\mathrm{gha\,ha^{-1}}$), eigenvalues of the deficit-side matrix J of (19) and the demand ceiling, for a sweep of the fast-land yield b_f (all other parameters at baseline). (b) Initial population P_0 (cap; initial footprint $E_0 = 0.55P_0$ in gha) at which the minimum capital land reached during an endogenous-demand run ($A_{f0} = 0.6\,\mathrm{ha}$, dimensionless $q_0 = 0.9$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$) comes within the stated distance of the typed floor $A_c^{\min} = 0.05$; the first two columns are read from a P_0 grid of step 0.1 ($\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$), the last is a bisection using the 10^{-6} ha floor-proximity criterion ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$). Drawdown depth is continuous in P_0 (Fig. 1c); the thresholds are a band, not a single value.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\mathrm{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial demand P_0 at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

(iii) the degradation loss exceeds the maximal technology gain once the sustained shortfall exceeds $S^{\dagger} = (\eta/\alpha)\ln(1 + \Delta b/b_{f0})$*, equal to* 1.30 gha *at the baseline.*

Under an additive degradation law $b_f = b_{f0} + T_b - \alpha D$ *the two channels enter symmetrically and (iii) reduces to a comparison of* Δb *with* αD^**; the asymmetry is a property of the multiplicative form.*

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S} = \sup_{s \le t} [E - B]_+$, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ by comparison; if $[E - B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1 + \Delta b/b_{f0})$. □

Under fixed demand and a sustained shortfall, D converges to S^*/η; the equilibrium increases with the shortfall. At baseline, the crossover shortfall $S^{\dagger} \approx 1.3$ gha exceeds baseline biocapacity B, so the floor-balance mechanism in Proposition 1(ii) bounds the mask while debt contributes a slower reduction in fast-land yield. A technology-induced increase in B also raises carrying capacity and, after the demographic delay, can increase demand. Its effect on debt peaks depends on initial conditions and timing. In the matched experiment in SI §S7, the all-time peak over $0 \le t \le 250$ yr at $P_0 = 1.0$ cap is 0.02020 gha yr with the wave and 0.00813 gha yr without it (ratio 2.49). At $P_0 = 1.2$ cap, the all-time peak is lower with the wave (0.05104 versus 0.06122 gha yr), while the post-midpoint peak over $t \ge t_{wave} = 40$ yr is higher (0.02518 versus 0.01085 gha yr; ratio 2.32). These results show that the wave's effect differs between the all-time and post-midpoint summaries and across initial conditions. The delayed debt rebound is conditional on the specified parameters and is neither a forecast nor an empirical estimate.

3.4. Recovery requires an explicit, gated restoration flow

The model permits conversion from capital land to fast land, but includes no reverse transfer except the restoration flow R_{rc}. Fast-land retirement transfers area to the reserve, not to capital land. Lowering population, footprint, or shortfall therefore does not reverse prior conversion; when $\chi_r = 0$, A_c remains at the post-conversion level. In the endogenous-demand run with $A_{c0} = 1.0$ ha and $P_0 = 2.0$ cap, A_c falls to 0.398 ha within four years and remains there without restoration. With surplus-gated restoration ($\chi_r = 0.10\,\text{gha}^{-1}\,\text{yr}^{-1}$), it recovers to 0.828 ha, corresponding to $\mathcal{R}_c = (A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}}) = 0.71$ (SI §S4.3). The gate sign determines when restoration acts. Surplus-gated restoration is inactive during a

shortfall; after a surplus opens, it can continue as population re-equilibrates, so restoration to a specified target requires a target-limited rule (SI §S4.4). Deficit-gated restoration acts during the shortfall but removes productive area while provision is insufficient; at the tested rates, conversion exceeds restoration and drawdown continues. The effects of both operators depend on their timing and rates.

3.5. NFA account composition, 1961–2023

The primary empirical window is the complete world series in NFA Edition 2026, results and metadata v1.0; 2024–2025 estimated rows are excluded. The 1961–2022 sensitivity uses the same edition and population series. Total biocapacity account value B rises from 9.7254×10^9 to 12.1428×10^9 gha (+24.86%, $\Delta \ln B = +0.221997$). Population from the workbook's UN PopStat field rises from 3.06487 to 8.09173 billion. These totals imply B/P falls from 3.17319 to 1.50065 gha person^{-1} (−52.71%, $\Delta \ln(B/P) = -0.748840$), while the Global Ecological Footprint of Production per person rises from 2.19700 to 2.56462 gha person^{-1} (+16.73%). The production-footprint-to-biocapacity ratio E_{prod}/B rises from 0.69236 to 1.70901 and first exceeds one in 1973. The exact identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ gives $+0.970837 - 0.748840 = +0.221997$; this is an extensive/intensive accounting split, not a causal decomposition. The same-edition 1961–2022 sensitivity gives $\Delta \ln B = +0.219497$, a 52.41% fall in B/P, a 17.70% rise in E_{prod}/P, and the same first-crossing year (SI §S5).

The land-type components make the NFA account composition explicit (Fig. 3; Table 3). Cropland biocapacity contributes +24.06 percentage points to the +24.86% aggregate change, about 96.8% of the net percentage change. Forest biocapacity contributes −5.95 points, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum, which includes built-up land, is nearly flat (1.0092×, or +0.80 points) only because it nets the built-up increase against declines in other categories. This is measured composition of NFA account values. It does not establish that forest biomass, soil condition, wetland extent, fish biomass, or the model's A_c declined; built-up land is an accounting class, not ecological capital. Component values sum to total biocapacity within workbook rounding at both endpoints.

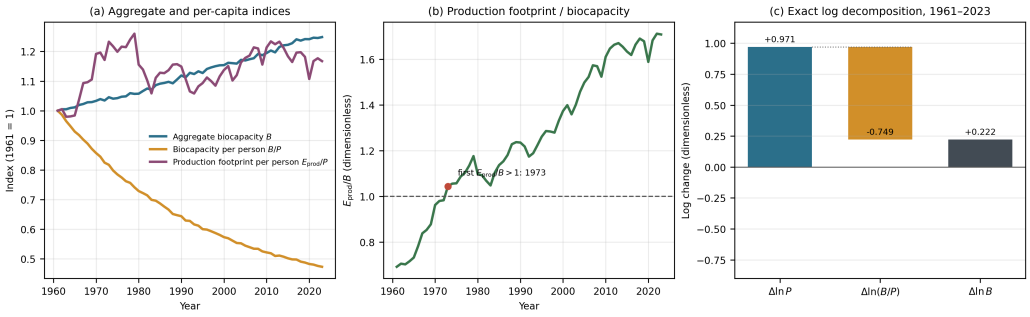


Figure 2: Aggregate and per-capita movements in NFA Edition 2026 (world, 1961–2023; complete observations). B is total biocapacity, E_{prod} is the workbook's Global Ecological Footprint of Production, and P is the single UN PopStat population series; per-capita values are calculated from totals. (a) Indices (1961 = 1) of B, B/P, and E_{prod}/P. (b) E_{prod}/B, first above one in 1973. (c) Exact log identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$. Account values are in gha; plotted indices, ratios, and log changes are dimensionless. The 2024–2025 estimated rows are excluded.

Table 3: NFA Edition 2026 biocapacity-account endpoint ratios and contributions. Contributions are $100\,[B_i(T) - B_i(1961)]/B(1961)$ percentage points; endpoint ratios are $B_i(T)/B_i(1961)$. The 2022 sensitivity is from the same edition. All entries are account values, not physical-stock measurements.

Component	2023 ratio	2023 contribution (pp of B_{1961})	2022 sensitivity ratio	2022 contribution (pp of B_{1961})
Cropland	2.8701	+24.0577	2.8514	+23.8183
Grazing land	0.9828	−0.2688	0.9815	−0.2890
Forest biocapacity	0.8996	−5.9525	0.9019	−5.8137
Fishing grounds	0.9627	−0.4212	0.9596	−0.4559
Built-up land†	8.7935	+7.4416	8.6298	+7.2854
Non-cropland sum†	1.0092	+0.7990	1.0083	+0.7267
Total biocapacity	1.2486	+24.8568	1.2454	+24.5450

†Built-up land is not ecological capital; the non-cropland account sum includes it. Component sums match total biocapacity within workbook rounding.

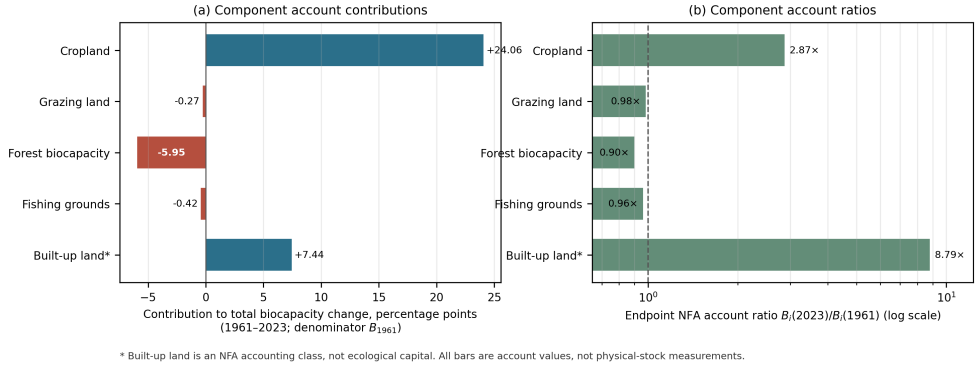


Figure 3: NFA Edition 2026 land-type biocapacity-account changes, world, 1961–2023. (a) Component contributions in percentage points of B_{1961}. (b) Endpoint ratios $B_i(2023)/B_i(1961)$. "Forest biocapacity" is the workbook's exact label. All bars are account values, not measurements of physical stocks; built-up land is an NFA accounting class, not ecological capital.

3.6. Falsifiable predictions

The model yields the following testable predictions. Tests of land-class composition require observations of aggregate biocapacity, capital-land area, fast-land area, and relevant land-cover or yield proxies, rather than aggregate biocapacity alone.

P1. **Mask signature.** In the baseline frozen-demand run, the model predicts a finite interval with $\dot{A}_c < 0$ and $\dot{B} > 0$ (Fig. 1a). The conversion contribution is positive exactly when $\Delta b_{\text{conv}} > 0$, but the net sign of $\dot{B}$ also depends on maintenance/retirement, quality growth, debt, technology, and restoration. With baseline shares $\sigma_f = \sigma_c = 1$, active conversion requires $R_B = E/B > 1$; in the reserved-flow extension with $\sigma_f, \sigma_c < 1$, conversion can occur even when $R_B \leq 1$, which is not a baseline result. Joint observations of B, E, and A_c distinguish the net composition signature (B rises while A_c falls) from a trajectory with stable E and declining A_c (Proposition 1).

P2. **Window scale.** The duration of the mask is set by the slow eigenvalue λ_1 (through τ_{conv}, μ, η_f, b_c/b_f) and its amplitude by Δb_{conv} and the deficit, not by a fixed horizon; the bulk of conversion precedes the closing of the deficit by one to two e-folding times.

P3. **No critical-slowing precursor.** The return time does not grow as demand approaches the ceiling, and the drawdown depth is continuous in demand (Fig. 1c); the model signal is the approach of A_c to its typed floor (Proposition 2); a fold-type early-warning signal is not predicted for this floor collision.

P4. **Demand threshold, not delay threshold.** The operational initial-demand threshold for approaching the capital floor rises with initial capital (Table 2b) and is insensitive to the regeneration delay (identical to 10^{-3} for $\tau_g = 5$–$40\,\mathrm{yr}$), consistent with the modest regenerative contribution and monotone quality dynamics in the baseline.

P5. **Recovery requires a gated flow.** Reducing demand alone does not restore capital-land area; recovery requires an explicit restoration flow; a surplus-gated flow is inactive during shortfall, whereas a deficit-gated flow can deepen the provision deficit.

The predictions cannot be tested from aggregate B alone. Tests of the composition mechanism require independent observations or defensible proxies for A_c and fast-land yield b_f; tests of the delay, recovery, and demand-threshold predictions require corresponding trajectories and intervention or demographic data.

4. Discussion

4.1. Structural implications of land-class heterogeneity

The two-land model yields three results relevant to aggregate representations of human–environment systems (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees, 1996; Brander and Taylor, 1998). First, conversion can increase aggregate biocapacity while capital-land area A_c declines, although the conversion contribution alone does not determine the net change in B; the exact sign condition is given by Proposition 1, and Proposition 3 bounds the technology/debt mechanism. Second, under the reduced frozen-demand assumptions of Proposition 2 ($q = q_{\max}$, $D = 0$, technology wave off, and restoration disabled), the balance relation is strictly monotone and its collision with the typed floor is not a fold. The absence of hysteresis and critical slowing down applies to this reduced equilibrium relation, not to the history dependence of realized states. With conversion but no return flow, A_c remains below its pre-overshoot value after conversion

stops, while $\dot{A}_c = 0$ at equilibrium (Table 4). Third, recovery requires an explicit restoration flow, whose gate has phase-dependent consequences.

Table 4: Equilibrium and trajectory properties under the stated assumptions.

Property	Finding	Scope
Fold on the reduced frozen-demand branch	Absent	$B_{eq}(A_c)$ is strictly monotone; the balance point collides with the typed floor at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing as $E \to E_{\text{ceil}}$	Absent	The slow eigenvalue remains negative and constant along the reduced branch
Hysteresis of the frozen-demand equilibrium relation	Absent	There is at most one feasible balance point; this does not imply history independence of realized states
Path dependence of realized A_c	Present, conditional	With $R_{rc} = 0$, one-way conversion leaves A_c below its pre-overshoot level after shortfall closure, while $\dot{A}_c = 0$ at equilibrium
Discontinuous response to endogenous demand	Not observed	In the reported runs, minimum A_c varies continuously and monotonically with initial population; proximity thresholds depend on tolerance

These equilibrium results qualify generic expectations for early-warning indicators. Rising autocorrelation and variance are associated with approach to a fold (Scheffer et al., 2001). When drawdown instead occurs through a typed-floor collision, such indicators need not rise before the floor is approached. Their absence is not evidence of safety; the relevant signal is the trajectory of A_c, which is not identified by the aggregate alone.

4.2. Measurement implications

Biocapacity in gha is a value-weighted scalar and does not identify the composition of its underlying land classes. Proposition 1(iii) formalizes this limit: at fixed per-hectare yields, the aggregate supplies one constraint on two land areas, leaving a continuum of feasible decompositions. The aggregate is the linear, perfectly substitutable member of the aggregator family in related

assessment work (Lemma A of Abaee, 2026b); it can satisfy an aggregate floor while a typed floor on A_c is violated. In that framework, no positive-elasticity aggregator is uniformly safe (Theorem S2(ii) in Abaee (2026b)). Under pathwise feasibility, constraints apply throughout the transition, so an intermediate decline below a typed floor cannot be offset by recovery at the endpoint.

For the National Ecological Footprint and Biocapacity Accounts, these results concern account composition. In Edition 2026, world biocapacity account value B rises 24.86% over 1961–2023, while B/P falls 52.71%; the global Ecological Footprint of Production per person rises 16.73%, and E_{prod}/B first exceeds one in 1973. At the account-component level, cropland contributes +24.06 percentage points of the aggregate change, Forest biocapacity −5.95, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum is nearly flat only because it combines those movements, including the large built-up increase. The 1961–2022 same-edition sensitivity is reported separately (SI §S5). These are movements in NFA account values expressed in gha. They are not measurements of forest biomass or condition, soil carbon, wetland extent, fish biomass, or other physical stocks; the decline in Forest biocapacity is not evidence by itself of a measured physical-stock decline. The built-up component is an accounting class, not ecological capital. Component ratios, contributions, vintage, and endpoint sensitivity characterize the account composition without establishing a physical-stock change (Lo et al., 2026; Borucke et al., 2013; Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015).

The modeled trajectory is conditional and distinct from the NFA account results. In the frozen-demand baseline (Fig. 1a), A_c falls from 1.500 to 0.807659 ha (−46.16%) while model biocapacity B rises from 0.594 to 0.825 gha (+39%). The quality state q evolves separately from 0.9 to 1.0; because $q_0 > q^* = q_{\max}/2$, the per-hectare value $\Upsilon_c(q)$ falls from 0.05576 to 0.05000 gha ha^{-1} (−10.33%). Thus the modeled capital contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (−51.72%). With restoration disabled, one-way conversion leaves A_c below its pre-overshoot level after the shortfall closes; at the balance point conversion stops, so the lower area persists rather than declining perpetually. These are conditional model outputs, not observations or estimates of global physical-stock change.

A common-boundary global physical-stock series for the relevant components was not assembled. Accordingly, no physical-stock composite or

global decline finding is reported. The SI scoreboard specifies candidate stock, extent, condition, and annual-flow measures together with comparability requirements; it contains no empirical endpoint values. Heterogeneous measures remain a nonadditive vector unless common boundaries, units, uncertainty, overlap, valuation or normalization, and permitted substitution are specified. Ecosystem extent, condition, services, and monetary presentations are distinguished in SEEA Ecosystem Accounting (United Nations et al., 2021).

The external cropland-area and cereal-yield proxy is an illustrative observation operator with explicit calibration and boundary limitations. Its 1961 anchor is the NFA cropland account share; cereal yield is a sentinel rather than an all-crops measure, and the physical-area and account boundaries are not identical by construction. The residual $C = B - \hat{b}_f A_f$ can therefore include calibration and boundary mismatch, other account components, and conversion-factor changes. It is not the model contribution $\Upsilon_c(q)A_c$ or a physical-capital measure. A physical interpretation would require independent, boundary-consistent observations of area, condition, yield, and regeneration, together with a validated mapping and valuation. Forest area may inform one component of the effective capital class but is not a complete measure of that state or of heterogeneous ecological assets.

4.3. Policy operators

The typed-floor analysis distinguishes three policy operators. A conversion stop triggered directly by the typed floor $A_c = A_c^{\min}$ guarantees that constraint. An A_f ceiling does not by itself guarantee a specified capital-land floor unless its stop rule is linked to the floor and remaining-area constraints. A governor that observes only aggregate B cannot guarantee the capital-land floor because B does not identify A_c. Surplus-gated restoration is inactive during shortfall, whereas a deficit-gated flow acts during it and can reduce provision further. These gate-specific outcomes are consistent with the distinction between protective and mobilizing institutional feedback in related work (Abaee, 2026a). With a common reserved share $\sigma_f = \sigma_c = \sigma < 1$, conversion is inactive on a trajectory while $E \leq \sigma B$; with distinct shares the corresponding condition is $E \leq \sigma_f Y_f + \sigma_c Y_c$. Robustness to disturbances and parameter uncertainty is not established.

4.4. Relation to related work

Related work addresses typed material-flow ledgers and compensatory aggregation (Abaee, 2026b,c), and distinguishes mobilizing from protective institutional feedback (Abaee, 2026a). The present model focuses on ecological-regeneration and demographic delays; institutional review delays are not included.

4.5. Limitations

Ecological-capital land aggregates heterogeneous assets, so its single regeneration state, composition diagnostic, and demand ceiling are effective model quantities. Most parameters are representative rather than estimated. The values b_c, $b_{G,c}$, and ρ_c are not separately identified by a cross-section; recovery observations span different systems and provide timescale context rather than a calibration of ρ_c; they concern regeneration response, not the lag τ_g. The per-capita footprint is held constant to isolate stock–flow–demand feedbacks; endogenizing e is an extension. The composition window is demand-driven and bounded under the stated mechanisms. Stability at $S = 0$ is one-sided because the positive-part operator is exact. Drawdown thresholds depend on the floor-proximity tolerance and are reported as bands, not sharp boundaries. The debt-repayment rate is fixed at $\eta = 0.05\,\mathrm{yr}^{-1}$; the limit $\eta \to 0$ removes the debt equilibrium and is excluded. NFA accounts do not separate physical area from yield, so they cannot identify those components independently. Under current-quality regeneration $G_c(q(t))$, the demonstration mask window is 1.3 yr longer and the initial-population threshold shifts by at most 0.001 cap relative to the delayed specification $G_c(q(t - \tau_g))$ (SI §S3).

5. Conclusions

Aggregate biocapacity does not identify its composition. In the two-land model, conversion contributes positively to B when fast-land yield exceeds the effective per-hectare value of capital land, while A_c declines; the net change in B depends on the other terms (Proposition 1). Under the stated reduced frozen-demand assumptions, the equilibrium relation is single-valued and has no fold, hysteresis, or critical slowing down. The realized trajectory is nevertheless path-dependent: with restoration disabled, one-way conversion leaves A_c below its pre-overshoot value after the shortfall closes, while $\dot{A}_c = 0$ at the balance point. In the representative run, A_c falls 46.16%, $\Upsilon_c(q)$

falls 10.33%, and $\Upsilon_c(q)A_c$ falls 51.72% while model B rises 39%. These are conditional model outputs, not empirical global-stock estimates.

The Edition 2026 NFA world accounts show that from 1961 to 2023 total B rises 24.86%, B/P falls 52.71%, and the global production footprint per person rises 16.73%. Cropland contributes +24.06 percentage points to the total biocapacity-account change; Forest biocapacity contributes −5.95 points, and built-up land contributes +7.44 points as an accounting class, not ecological capital. These results describe account composition, not changes in biomass, soil condition, wetland extent, fish biomass, or A_c. The 1961–2022 sensitivity uses the same NFA edition and population series. A common-boundary global physical-stock series for the relevant components was not assembled; accordingly, no physical-stock composite or global decline finding is reported. The SI scoreboard remains an unpopulated specification, and heterogeneous physical-stock measures remain a nonadditive vector absent common spatial and temporal boundaries, normalization or valuation, and explicit substitution rules. The proxy residual $C = B - \hat{b}_f A_f$ is neither a physical-capital measure nor the model contribution $\Upsilon_c(q)A_c$. Physical interpretation requires independent, boundary-consistent observations and a validated mapping.

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Declarations

Data availability. All empirical inputs are publicly available from the sources cited in the manuscript. The NFA Edition 2026 workbook and release notes are available from York University's Ecological Footprint Initiative; the cropland-area and cereal-yield series are available from the Our World in Data pages cited in the references. Derived analysis data and model code are available from the corresponding author upon reasonable request.

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